

Schema Change Request

Columns, behaviour changes, and defects

VERSION	1.1
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Status	For review. Reconciled against the 2026-08-23 snapshot

Everything the audit engine needs added to the Cartograph schema, with the feature each change enables.

Verified against the repository at the 2026-08-23 snapshot (model validations, strong-parameter lists, and 456 migrations), so nothing here duplicates a column that already exists. Items satisfied since v1.0 have been removed.

Organised in three tiers: **behaviour changes** (Section 1) that alter how the importer works and carry consequences if omitted; **new columns** (Section 2) that enable specific features; and **defects** (Section 3) found during review that apply regardless of this integration.

1. Behaviour changes

These are not column additions. Each changes how something works, and each has a consequence attached.

1.1 source column on the cert child tables

Timing: not needed for the first import. Required before the second.

TABLE	COLUMN	TYPE	DEFAULT
cert_household_members	source	string	"manual"
cert_income_records	source	string	"manual"
cert_asset_records	source	string	"manual"
cert_expense_records	source	string	"manual"
cert_review_findings	source	string	"manual"

Why. Re-import clears extraction-owned child records and rewrites them. Without a marker distinguishing what the importer created from what an analyst typed by hand, the delete removes both. The failure appears only on the second import of a case, by which point reviewer work is already in place.

With the column, the delete becomes `where(source: "extraction")`. Existing rows default to "manual" and are never touched. The change is additive, with no behaviour change for anything already in the database.

Until this exists, the importer must treat a cert review that already has records as 409 rather than proceeding.

1.2 AR-SC in CERT_TYPES

CertReview::CERT_TYPES currently allows initial, annual, interim. The engine audits four certification types: MI, AR, **AR-SC**, and IR. Three map cleanly; AR-SC has no target.

Either add a fourth value, or agree that AR-SC maps to annual with the distinction carried in another field. The engine applies different rules to AR-SC, which was specific development work, so collapsing it silently would lose a real distinction.

1.3 Finding reference on correction uploads

JobDocument.document_class includes corrections, which tells the engine that a file **is** a correction but not **which finding** it answers.

A cert_review_finding_id (nullable, indexed) on job_documents, or an equivalent parameter on the client upload path, enables the targeted verification flow: the engine verifies one document against one finding rather than re-auditing the whole packet. A targeted verification costs roughly 2 to 5 cents and completes in seconds, against approximately 60 cents and several minutes for a full re-audit. It is also the mechanism behind auto-approving a resolved item.

1.4 item_code on cert_review_findings

This is the single highest-priority item in this document.

TABLE	COLUMN	TYPE	NOTES
cert_review_findings	item_code	string, indexed	Nullable. Null for findings not tied to a checklist item

item_code was added to cert_checklist_templates on 2026-08-18, and cert_review_checklist_items was dropped the same day. That leaves the template holding item definitions and the finding holding per-case results, with **nothing connecting them**: a finding has no way to reference the item it answers.

Until this column exists, the engine can determine that a checklist item passes or fails but has nowhere to record it, and the checklist cannot render pre-answered. ProcessRunpod0crScanJob's own TODO refers to pre-answering CertReviewChecklistItem rows, which no longer exist.

Also needed, and not a code change: the item_code **values for each program**. They are editable per community funding program, so the engine cannot hardcode them.

2. New columns

2.1 cert_review_findings (highest-value additions)

COLUMN	TYPE	ENABLES
item_code	string, indexed	See Section 1.4. Links a finding to the checklist item it answers
finding_key	string, indexed	Stable identity across re-imports, so reviewer resolutions survive and findings do not duplicate

COLUMN	TYPE	ENABLES
confidence	decimal(4,3)	Auto-approve gating. A determination's confidence, alongside its nature, is what decides whether an item can clear without manual review
resolution_type	string	presence_only or recalculation. Tells Cartograph whether resolving this finding is a document check or triggers a recalculation, which is the other half of the auto-approve decision
pages	integer array or JSON	Reviewer guidance. Lets the UI link to the pages the finding came from, for example "see pages 12 to 14"
detail	text	Case-specific reviewer guidance, internal only, never client-visible
source_document_id	bigint, nullable	Which uploaded document the finding came from, for page links to resolve

finding_key, confidence, and resolution_type block features. pages and detail are required for reviewer guidance.

Constraint on confidence. The score currently blends extraction quality with agreement against the MuleSoft extraction. When MuleSoft is retired, that second component loses its counterparty and the score requires rebuilding and recalibration. The column can be added now; the thresholds that gate auto-approval should be set once verified-case data is available to support them.

2.2 cert_reviews (previous-certification comparison)

COLUMN	TYPE	ENABLES
prior_effective_date	date	AR and IR comparison. The engine extracts the previous certification from the packet and currently has nowhere to put it
prior_annual_income	decimal(12,2)	Year-over-year income delta
prior_tenant_rent	decimal(12,2)	Rent change detection
prior_gross_rent	decimal(12,2)	
prior_utility_allowance	decimal(12,2)	
prior_household_size	integer	Household composition change
extraction_id	string	Provenance. Links a cert review back to the extraction that populated it

Interim recertifications are defined by what changed since the last certification. Without these columns the comparison has no destination, and the reviewer sees the new figures with no baseline.

2.3 cert_income_records (finding evidence)

COLUMN	TYPE	ENABLES
employment_status	string	Active / Terminated / On Leave. Drives terminated-employment and stale-wage findings
termination_date	date	Same

Without these, Cartograph can display *that* a finding fired but not the evidence behind it, which undercuts reviewer guidance.

2.4 cert_asset_records (completeness)

COLUMN	TYPE	ENABLES
account_number	string	Matching an asset to its statements; detecting duplicate disclosures
interest_type	string	Distinguishes actual from imputed asset income

Lower priority than the preceding sections: useful detail rather than blocked features.

2.5 cert_asset_bank_statements (completeness)

COLUMN	TYPE	ENABLES
account_number	string	Matching statements to the right account when a household holds several

3. Defects found during review

Independent of this integration, and each verified against the migration history.

3.1 Three permitted parameters have no backing column

Records are built by mass assignment, so including any of these raises `ActiveModel::UnknownAttributeError` and returns a 500.

- `cert_asset_record[self_declared_balance]`: no column. `self_declared_balance` is a seeded calculation-method code. The real columns are `self_declared_income` and `manual_balance`.
- `cert_expense_record[monthly_amount]`: no column.
- `cert_expense_record[verification_status]`: no column. Income and asset records have one, expenses do not.

3.2 The Salesforce mapper emits values the validators reject

map_income_type produces zero_income, unemployment, and disability; map_asset_type produces cash and life_insurance. None appear in INCOME_TYPES or ASSET_TYPES. Assets are created with create!, so those raise RecordInvalid. Note unemployment versus the valid unemployment_benefits in particular. The fallback path also slugifies arbitrary input, which can emit anything.

3.3 SSN is stored in plaintext

CertHouseholdMember#ssn is a plain text column with no encrypts declaration anywhere in the model layer. Until that changes, the audit engine sends ssn_last4 only and never full digits. Its own store keeps SSNs masked at every egress point, and writing them into an unencrypted column would be a downgrade in handling.

Adding encrypts :ssn plus a backfill would allow full write-back later if it is ever needed.

3.4 Webhook signature verification fails open

The three existing webhook controllers proceed when their signing secret environment variable is unset: Stripe parses unsigned JSON, Resend logs a warning and continues, Auth0 skips the check. The new ingest endpoint should return 401 instead, and reject any request whose timestamp is more than 300 seconds old.

3.5 Paystub and VOI data have two homes

cert_income_records carries JSONB columns sf_paystubs, sf_voi_records, and sf_zero_income_worksheets, written by the Salesforce import via update_columns (which skips validation), alongside the real associated tables written by the UI. CertIncomeCalculationService reads the relational tables. The engine targets the relational tables; the JSONB columns appear to be a migration artifact and are candidates for retirement.

4. Summary by priority

PRIORITY	ITEM	BLOCKS
1	item_code column on cert_review_findings	Checklist pre-answering. Nothing currently links a determination to an item
1	item_code values per program *(export, not a code change)*	Same. Codes are editable per program, so they cannot be assumed
2	source column on five tables	Safe re-import; data loss without it
3	finding_key, confidence, resolution_type on findings	Auto-approve, resolution survival
4	pages, detail, source_document_id on findings	Reviewer guidance
5	AR-SC decision	One of four certification types
6	Finding reference on correction uploads	Targeted correction verification

PRIORITY	ITEM	BLOCKS
7	Prior-certification columns on cert_reviews	AR and IR comparison
8	employment_status, termination_date	Finding evidence
9	Remaining completeness columns	Detail